AMS 572 Project Report

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Contents

1 Introduction

TODO (Jacob): Format bibliography w/ links to sources Nearly two decades since its introduction in 2009, Bitcoin has gained mainstream traction as an investment asset. This digital currency is held by major hedge funds[1], Fortune 500 companies[2], and national banks[3]. Additionally, Bitcoin has gained acceptance as an everyday currency with many retailers now accepting it as payment. Most incredibly, Bitcoin was even declared a legal-tender currency of El Salvador in September 2021[4]. As Bitcoin has become more accepted, retail investors calmored to buy and trade this new asset. Today, you can hold Bitcoin in digital wallets, investment and retirement accounts[5], as well as purchasing shares of Bitcoin-focused exchange-traded funds (ETFs)[6].

Moreover, the meteoric rise in price per bitcoin has spawned a new class of digital assets called cryptocurrencies. Aside from Bitcoin, these cryptocurrencies follow one of two paradigms. The first paradigm describes coins that focus on improving some aspects of the technology such as energy efficiency, transaction speed, or added utility. These coins utilize their own blockchain networking structure leading to their name as "native coins". The second type of coin attempts to capitalize on the speculation and excitement around Bitcoin's price by creating eye-catching imitations. These coins use existing blockchain networks and are often themed around a specific internet trend or personality to garner attention. In online circles, these types of coins have earned the title of "meme coin" as they make no novel improvements to the underlying structure.

Due to its mainstream adoption in traditional financial markets by both institutional and retail investors, does Bitcoin's price follow stock market movements? Is Bitcoin truly a new investment asset, or is its price so deeply entangled with traditional investments that it has lost its novelty? The answer to this question has serious ramifications for portfolio allocations. If Bitcoin's movements are unrelated to the stock market, it can be a valuable tool to diversify portfolios and reduce risk. However, if Bitcoin's price follows the movements of the stock market, its utility as an investment asset mirrors similar stocks in the digital sphere.

How do the price movements of Bitcoin proportionally affect Native or Meme coins? Are investors more excited about the underlying technology powering digital assets or are they seeking an avenue to wildly speculate? The answer to this question will help unravel the motivation fueling the cryptocurrency boom. This is because of the dual purpose of currency, being a store-of-value and a mode of facilitating transactions. Both of these purposes benefit from the price being predictable and stable. If the meme coins are more sensitive to changes in Bitcoin, it highlights their use as a speculative asset. Likewise, this result highlights the utility of native coins and investors bet on the underlying technological improvements.

1.1 Questions of Interest

How do the price movements of Bitcoin proportionally affect Native or Meme coins? Are investors more excited about the underlying technology powering digital assets or are they seeking an avenue to wildly speculate? The answer to this question will help unravel the motivation fueling the cryptocurrency boom. This is because of the dual purpose of currency, being a store-of-value and a mode of facilitating transactions. Both of these purposes benefit from the price being predictable and stable. If the meme coins are more sensitive to changes in Bitcoin, it highlights their use as a speculative asset. Likewise, this result highlights the utility of native coins and investors bet on the underlying technological improvements.

1.2 Description of the Data

The data for this analysis was imported from Yahoo Finance using the `quantmod` package in `R`. This data contains the date, daily opening price, daily high price, daily closing price, and daily volume for each asset specified. The range of dates collected for this analysis ranges from November 1, 2023 to November 18, 2025. This data was then transformed to isolate the log-daily returns over this time period. Additionally, a dummy variable was added to categorize between the native coins and the meme coins. When combining the SPY and BTC data, non-trading days such as holidays and weekends were removed.

2 Background

In this section, we review some basic mathematical machinery with regard to financial time series, and discuss the tools we will be using to perform statistical inference.

2.1 Introduction to Financial Time Series

By a time series we mean a sequence of observations taken at regular time intervals. More formally, a **time series** is simply a sequence $\{x_t\}_{t \in T}$ of real numbers x_t, for $T = \{t_0, t_1, t_2, t_3, \ldots, t_n\}$ some finite indexing set of time increments over a certain period[1]. In the case of finance, our sequence $\{x_t\}_{t \in T}$ (which we may write equivalently as x_t for brevity, where the time index T is implicitly understood) represents the evolution of the price of a particular asset over time. To our notation consistent, we denote the price of a particular asset at the end of a trading day t by P_t .

When we study the evolution of an asset price, it is often useful to measure changes in price by computing the daily **rate of return** $\frac{P_t}{P_{t-1}}$ or, more preferably, the **daily log return** time series r_t of P_t, which is given by the formula

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right). \tag{1}$$

An example of a financial time series (i.e., the daily return of the S&P 500 stock market index), its daily return & daily log return time series is shown in Figure 1 below.

[1]In the case of real-world data, this indexing set is finite. However, generally we can make T to be any finite, countably infinite, or even uncountably infinite subset of $\mathbb{R}$ (the real numbers).

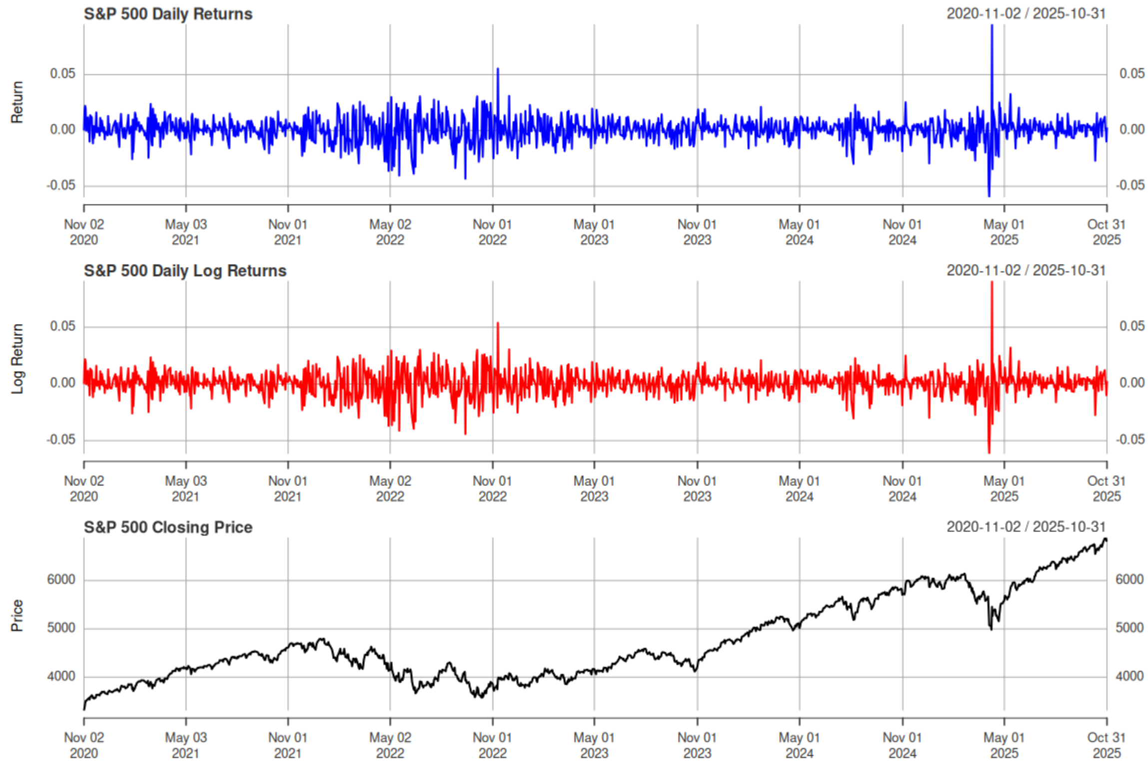

Figure 1: S&P 500 Daily Returns, Daily Log Returns, and Closing Prices between Nov 1, 2020 and Nov 1, 2025.

In what follows, we will use analyze the price of different cryptocurrencies and financial assets using log returns as in Equation 1.

2.2 Inference for the Correlation between Time Series

Recall that, if X and Y are continuous random variables with $E[X] = \mu_X$ and $E[Y] = \mu_Y$, variances $\text{Var}(X) = \sigma_X^2$ and $\text{Var}(Y) = \sigma_Y^2$, then their **covariance** is given by the formula

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$$

and subsequently their **correlation** is given by

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

How can we define the same notion of correlation for time series to asses a relationship between them? It turns out, if we are given two time series $\{x_t\}_{t=1}^N$ and $\{y_t\}_{t=1}^N$, we can use the **Pearson correlation coefficient**, defined in Equation 2 below as

$$\hat{\rho} = \frac{\sum_{t=1}^n (x_t - \overline{x})(y_t - \overline{y})}{\sqrt{\sum_{n=1}^n (x_t - \overline{x})^2}\sqrt{\sum_{t=1}^n (y_t - \overline{y})^2}}, \tag{2}$$

where $\overline{x}$ and $\overline{y}$ denote the sample means of $\{x_t\}_{t=1}^n$, $\{y_t\}_{t=1}^n$, respectively. I.e.,

$$\overline{x} = \frac{1}{n}\sum_{t=1}^n x_t, \qquad \overline{y} = \frac{1}{n}\sum_{t=1}^n y_t.$$

When studying the relationship between two time series x_t and y_t, we can use the Pearson correlation coefficient in Equation 2 to construct a hypothesis test of $H_0 : \hat{\rho} = 0$ vs. $H_1 : \hat{\rho} \neq 0$ under the following assumptions:

1. The pairs (x_t, y_t) are independent across time: for $t_1 \in T$ and $t_2 \in T$, (x_{t_1}, y_{t_1}) and (x_{t_2}, y_{t_2}) are independent, and
2. (x_t, y_t) are jointly normally distributed.

Under these assumptions, we use the test statistic

$$T = \hat{\rho}\sqrt{\frac{n-2}{1-\hat{\rho}^2}}, \tag{3}$$

where T follows a t-distribution with $df = n - 2$ degrees of freedom under H_0.

In practice, particularly when dealing with financial data, the assumptions 1 and 2 are often not well satisfied. For starters, the log return of any particular stock often *does not* follow a normal distribution, which violates assumption 2. Furthermore, the development of the price of a stock may not be generally independent across time. However, for purposes of simplification, it is often assumed to be the case in the theory financial modeling that stock prices are independently and identically normally distributed.

2.3 Linear Regression on Financial Time Series

For **Question 2**, we are interested in whether or not the price of memecoins track closer to the price of Bitcoin versus other, more serious nativecoins. We will study this question by performing a linear regression analysis. Let

$$\text{Memecoin}_i = \begin{cases} 1 & \text{if coin } i \text{ is a memecoin,} \\ 0 & \text{if coin } i \text{ is a nativecoin.} \end{cases}$$

We then fit the model

$$\text{Return}_{i,t} = \beta_0 + \beta_1 \text{BTC}_t + \beta_2 \text{Memecoin}_{i,t} + \beta_3 (\text{BTC}_t \times \text{Memecoin}_{i,t}) + \varepsilon_{i,t}. \tag{4}$$

We can interpret the coefficients in Equation 4 in the following way: β_1 can be viewed as the sensitivity of native coins to changes in the price of Bitcoin (notice that when $\text{Memecoin}_i = 0$ for a nativecoin i, the slope of Equation 4 is simply β_1). Moreover, β_3 can be viewed as the *difference* in sensitivity for memecoins versus nativecoins, since when $\text{Memecoin}_i = 1$ for a memecoin, the slope is $\beta_1 + \beta_3$. With this in mind, if it turns out that $\beta_3 > 0$, we have that memecoins are more sensitive to the price of Bitcoin than nativecoins are.

In order to apply the linear regression analysis as in Equation 4, which has the general form $Y = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ$, we require the following conditions to be satisfied by X, Z, and the interaction term XZ:

1. The conditional expectation of Y should be linear in the parameters, i.e., $\mathbb{E}[Y \mid X, Z] = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ$,
2. the regressors X, Z, and XZ should be uncorrelated with the error term,
3. no regressor is a linear combination of the others,
4. $\text{Var}(\varepsilon \mid X, Z)$ is constant (this is called **homoskedasticity**),

5. errors are uncorrelated across observations (this means that there is **no autocorrelation**), and

6. ε is normally distributed.

Again, in the case of financial time series, assumptions 4 and 5 are often not satisfied. As a result, if we were to perform a standard linear regression analysis, we would not be able to trust our OLS standard errors and their associated p-values. To remedy this fact, we instead compute a different flavor of standard errors, called **heteroskedasticity and autocorrelation consistent standard errors**, which will provide us with valid inference results even when assumptions 4 and 5 are not satisfied.

To motivate and illustrate this alternative approach, consider the standard linear model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

for $\mathbf{Y} \in \mathbb{R}^n$ a vector of responses, $\mathbf{X} \in \mathbb{R}^{n \times p}$ a design matrix, where the ith row is denoted X_i, $\boldsymbol{\beta} \in \mathbb{R}^p$ is the vector of parameters, and $\boldsymbol{\varepsilon} \in \mathbb{R}^n$ is the error vector, where under classical OLS assumptions we assume $\text{Var}(\varepsilon_i) = \sigma^2$, where σ is a **constant**, and $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$. We already know that the standard OLS estimator of $\boldsymbol{\beta}$ is given by

$$\widehat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{Y}$$

which is unbiased provided $[E](\boldsymbol{\varepsilon}) = \mathbf{0}$ and the errors are independent of $\mathbf{X}$. If we want to perform the hypothesis test

$$H_0 : \beta_j = 0,$$

and compute a p-value, we compute the test statistic

$$t_j = \frac{\widehat{\beta}_j}{\text{SE}(\widehat{\beta}_j)} = \frac{\widehat{\beta}_j}{\sqrt{\widehat{\text{Var}}(\widehat{\boldsymbol{\beta}})}} \tag{5}$$

where $\widehat{\mathrm{Var}}(\widehat{\boldsymbol{\beta}}) = \widehat{\sigma}^2(X^T X)^{-1}$ is the sample variance of the OLS estimator. However, the problem is, this formula for the sample variance is *only* valid when the assumption that the errors are IID holds. In the case where this does not hold (in particular, in the case where the errors are correlated and heteroskedastic), we need to use a different estimator of the covariance matrix of $\widehat{\boldsymbol{\beta}}$, called the ***Newey-West estimator***.

The Newey-West estimator of the covariance matrix of $\widehat{\boldsymbol{\beta}}$ is given by

$$\widehat{\mathrm{Var}}_{\mathrm{NW}}(\widehat{\boldsymbol{\beta}}) = (X^T X)^{-1} \left(X^T \widehat{\Omega}_{\mathrm{NW}} X\right) (X^T X)^{-1}, \tag{6}$$

where $\widehat{\Omega}_{\mathrm{NW}}$ is the complicated expression

$$\widehat{\Omega}_{\mathrm{NW}} = \left(\frac{1}{n}\sum_{t=1}^{n} \widehat{\varepsilon}_t \widehat{\varepsilon}_t \mathbf{X}_t \mathbf{X}_t^T\right) + \sum_{h=1}^{L}\left(1 - \frac{h}{L+1}\right)\left(\frac{1}{n}\sum_{t=h+1}^{n} \widehat{\varepsilon}_t \widehat{\varepsilon}_{t-h} \mathbf{X}_t \mathbf{X}_{t-h}^T + \left(\frac{1}{n}\sum_{t=h+1}^{n} \widehat{\varepsilon}_t \widehat{\varepsilon}_{t-h} \mathbf{X}_t \mathbf{X}_{t-h}^T\right)^T\right).$$

In this case, we replace $\mathrm{SE}(\widehat{\beta}_j)$ in Equation 5 with $\mathrm{SE}_{\mathrm{NW}} = \sqrt{\widehat{\mathrm{Var}}_{\mathrm{NW}}(\widehat{\boldsymbol{\beta}})}$ using Equation 6. The resulting p-values can then be trusted and the multiple linear regression analysis can be performed fully. In `R`, we simply use the `sandwich` and `lmtest` package, fit our linear model `model <- lm(...)` as usual, calculate the Newey-West covariance matrix with `NeweyWest(model)`, and display the regression results using `coeftest(model, NeweyWest(model))`.

2.4 Autocorrelation

In the previous section, it was mentioned that one of the crucial assumptions to apply linear regression analysis is that the errors are uncorrelated across observation, and in the specific case of time series, this is the requirement that there is no **autocorrelation**.

Generally speaking, the autocorrelation of a time series is a measurement of how much a particular time series of interest x_t correlates with a delayed copy of it x_s, where $s \neq t$ and $|t - s|$ is called the ***lag***. We define the ***autocorrelation function*** of a time series x_t as follows:

$$\rho_x(s,t) = \mathbb{E}[x_s \cdot x_t]. \tag{7}$$

Of course, in practice, we have to estimate ρ_x using sample data. In this case, given time series data $\{x_1, x_2, \dots, x_n\}$ (e.g., n days of closing prices), if we let $\overline{x} = \sum_{t=1}^{n} x_t/n$, and let

$$c_k := \sum_{t=1}^{n-k} \frac{(x_{t+k} - \overline{x})(x_t - \overline{x})}{n},$$

then the ***sample autocorrelation function*** is given by

$$\hat{\rho}_x(k) = \frac{c_k}{c_0}. \tag{8}$$

In this case, the input k is called the ***lag***. To assess whether or not a time series has autocorrelation, it is typically useful to plot values of $\hat{\rho}_x(k)$ for $k = 1, \dots, n$ in what is called a ***correlegram***. A correlegram plots the correlation between the time series and a copy of itself lagged by k in time.

3 Exploratory Analysis of the Data

To understand Bitcoin's behavior relative to a traditional equity benchmark we compare the growth of a $1 investment in BTC to the S&P 500 ETF (SPY), along with the corresponding daily volatility in Figure 2 below. These comparisons highlight the structural differences between cryptocurrency markets and conventional equity markets.

Figure 2: Growth of $1.00 USD invested in S&P 500 vs. Bitcoin from November 1, 2023 - November 18, 2025

To explore the relationship between daily returns of BTC with native coins and memecoins we started with scatterplots of individual coin returns against BTC along with their fitted linear regression lines. For native coins, we display this relationship in Figure 3 below.

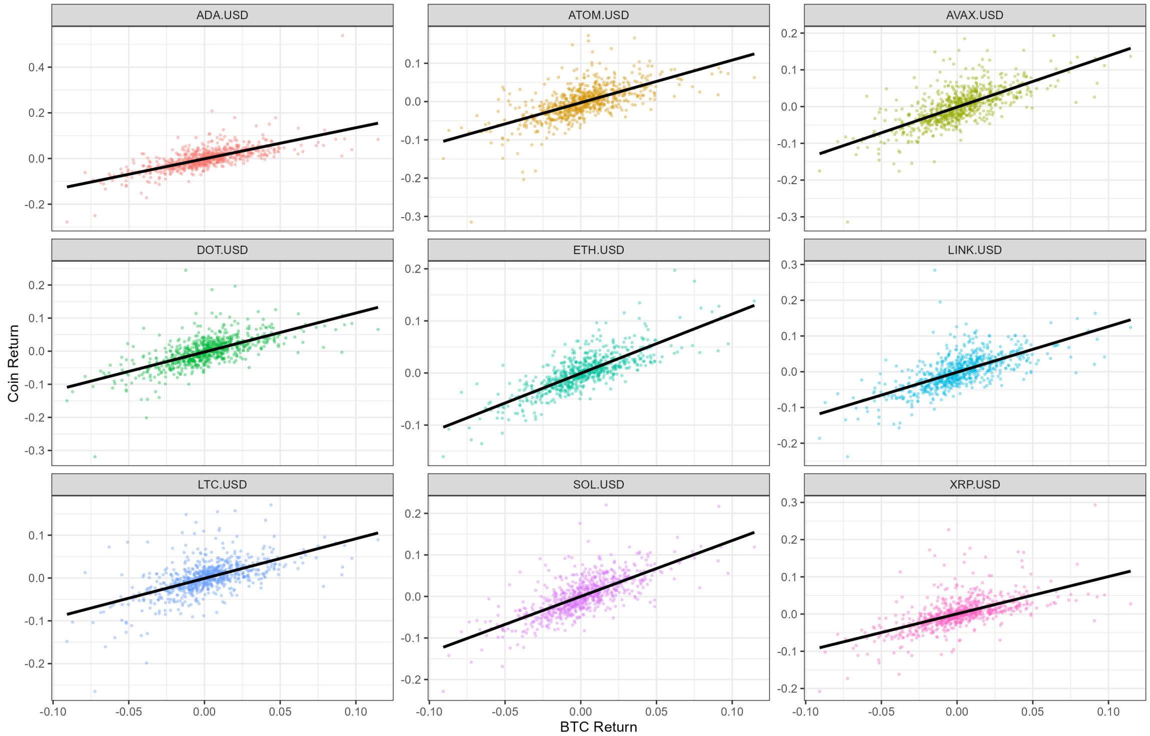

Figure 3: Scatter plots of individual native coin returns against BTC along with their fitted linear regression lines.

Across the native coins (ADA, ATOM, AVAX, DOT, ETH, LINK, LTC, SOL, XRP), the returns show a strong positive correlation with BTC. The scatterplots show relatively tight clustering around the fitted regression lines, and the slopes are uniformly positive. This suggests that native coins respond in a similar way to market movements of BTC. For memecoins, we display their relationship with BTC in Figure 4 below.

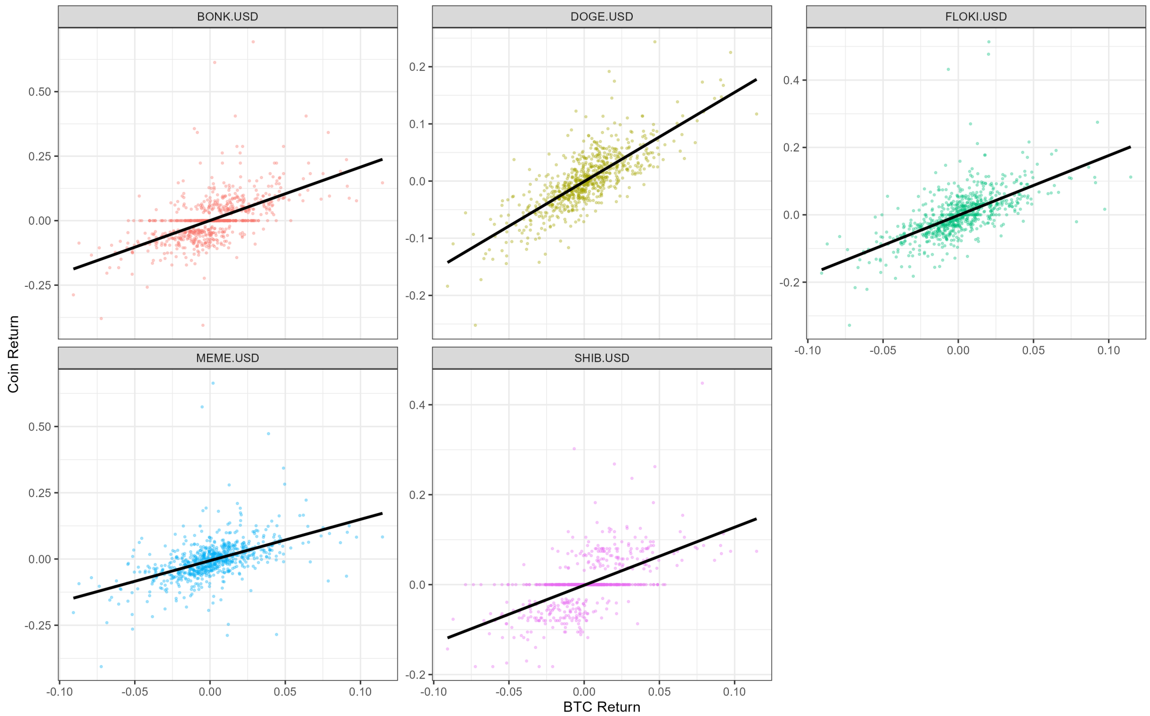

Figure 4: Scatter plots of individual memecoin returns against BTC along with their fitted linear regression lines.

The memecoin group (BONK, DOGE, FLOKI, MEME, SHIB) displays noticeably higher dispersion in returns and weaker alignment with BTC movements. While the fitted slopes remain positive for all tokens, the magnitude of the slopes is generally smaller than that observed for native coins, and the returns are more scattered. This suggests that memecoin prices are less attached to the broader crypto-market factor represented by BTC. These tokens appear to exhibit higher volatility. This volatility is often driven by community activity, speculative flows, and social-media-induced demand surges. Even where some memecoins (DOGE) show moderately strong BTC sensitivity, the noise around the fitted line remains substantially larger relative to native coins.

Overall this leads us to the idea that native coins exhibit stronger, more stable, and more linear sensitivity to BTC, whereas meme coins demonstrate weaker systematic exposure and substantially greater volatility. Native coins behave more like assets grounded in platform fundamentals, while memecoins resemble speculative instruments whose return profiles are heavily influenced by other speculative factors rather than market-wide factors. This motivates our deeper dive in question 2.

4 Analysis & Results

4.1 Question 1

Recall that our first question of study is the following:

Question 1: ***Does Bitcoin's price movements follow those of the S&P 500***?

To formally study this question, we set up the following hypothesis test in Equation 9 below using **Pearson's correlation coefficient** from Equation 2:

$$\boxed{H_0 : \hat{\rho} = 0 \qquad \text{vs.} \qquad H_1 : \hat{\rho} \neq 0} \tag{9}$$

The plot in Figure 5 below assesses linearity and is meant to show that model errors have a mean of zero.

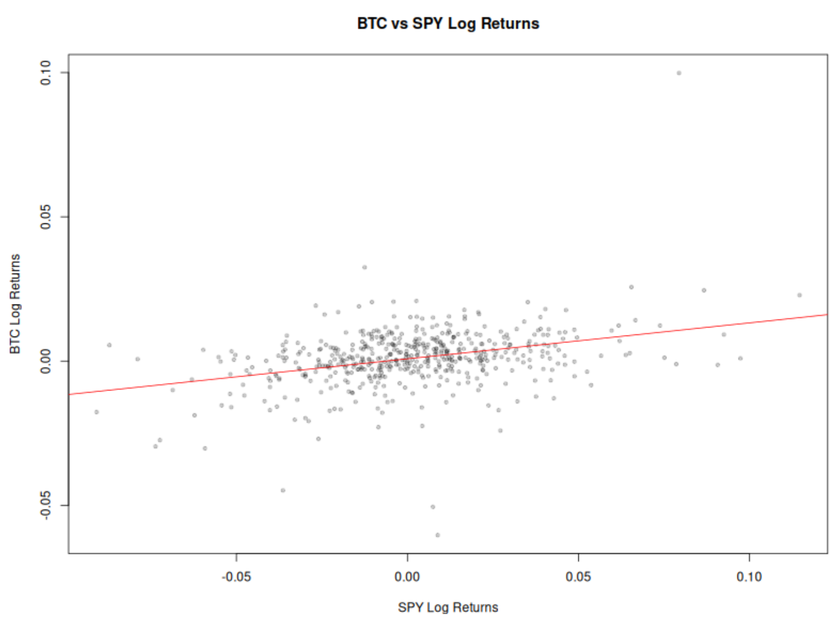

Figure 5: Plot of BTC vs. SPY returns with regression line. (Correlation coefficient: $\hat{\rho} \approx 0.343$.)

In order to perform this hypothesis test, we discuss the fulfillment of key assumptions:

1. **The data is generated from a normal distribution**. This assumption is likely to be violated. This is because financial returns often exhibit fat-tails. To mitigate this behavior of the data, we use the log-returns. This transformation should limit the excess kurtosis. Additionally, since log-returns are additive, the daily log-return is equivalent to the sum of many momentary returns. Due to the efficient market hypothesis[2], these returns result from a random walk. By the Central Limit Theorem, this sum should be approximately normally distributed especially when dealing with many data points. This assumption is visually assessed by examining the individual variables' QQ-plots.

2. **There must be a linear relationship between groups.** This again is a reasonable assumption as they are both heavily intertwined with our financial markets. When investors are optimistic about the future, one would expect both the SPY fund and the price of Bitcoin to move. Likewise, when investors are fearful about economic instability, money is generally reallocated away from assets with prospective growth towards assets with current earnings and utility. This assumption can be visually inspected using the scatterplot of the log-returns.

3. **The data is random and from independent samples.** For this time-series data, this means that there is no autocorrelation. This is a reasonable assumption as, due to the efficient market hypothesis, any obvious information from past data, is priced-in to the current valuation. This assumption is checked using an ACF plot to see if there is significant autocorrelation between the current log-return and lagging log-returns.

The residuals are clustered around zero implying that modeling log returns can be done using a linear approximation. There are noticeable outliers at the extreme fitter values of -.2 and .2. This

[2]The ***efficient market hypothesis*** says, in principle, says that one cannot consistently beat the market. In particular, the price of a particular asset reflects all available information, so that only new information moves prices. Since new information is unpredictable, it follows that price movements based on that information are unpredictable. Informally speaking, when we sum unpredictable events, we obtain a normally distributed ***random walk***. Therefore, the log return $r_t = \log(P_t/P_{t-1})$, which due to the property of logs can be written as the sum of unpredictable price changes $\log(P_t/P_{t-1}) = \sum_{s=1}^{t} \log(P_s/P_{s-1})$, can be viewed as itself a random walk, which is normally distributed.

is expected in crypto data since they are a volatile asset. These extreme points perhaps suggest that some coins or days had large returns relative to BTC.

The Q–Q plots for SPY and BTC log returns in Figure 6 below reveal important differences and similarities in their distributions relative to a normal distribution. For both assets, the mid-quantiles lie close to the theoretical line. This suggests the bulk of the return distribution is approximately symmetric and only moderately non-normal. However it is apparent that the points deviate from the straight line in both the lower and upper extremes. This indicates that both BTC and SPY returns are more heavy-tailed than a Normal distribution.

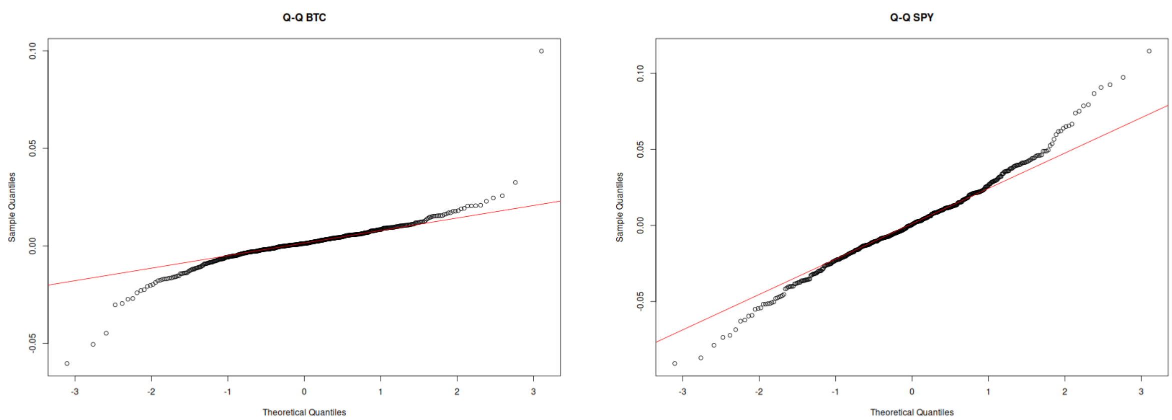

Figure 6: **Left**: Q-Q Plot of BTC returns. **Right**: Q-Q plot of SPY returns.

The BTC Q–Q plot shows a bit more curvature at the extremes, especially on the right tail, where several points rise slightly further from the line. There are one or two returns that stand further above the line than any corresponding point in SPY's Q–Q plot. These are individual observations and do represent a fundamentally different distributional shape

Overall, the Q–Q plots for BTC and SPY look quite similar, showing the typical heavy-tailed structure of financial returns. Both assets depart from normality in broadly the same way. While Bitcoin exhibits slightly stronger tail deviations the difference is modest. This small difference could imply that BTC is slightly more volatile than SPY.

4.1.1 The Effect of Missing Values for Question 1

4.2 Question 2

Recall that our first question of study is the following:

Question 2: ***Do memecoins follow the price changes of Bitcoin more than native coins?***

4.2.1 The Effect of Missing Values for Question 1

5 Conclusion